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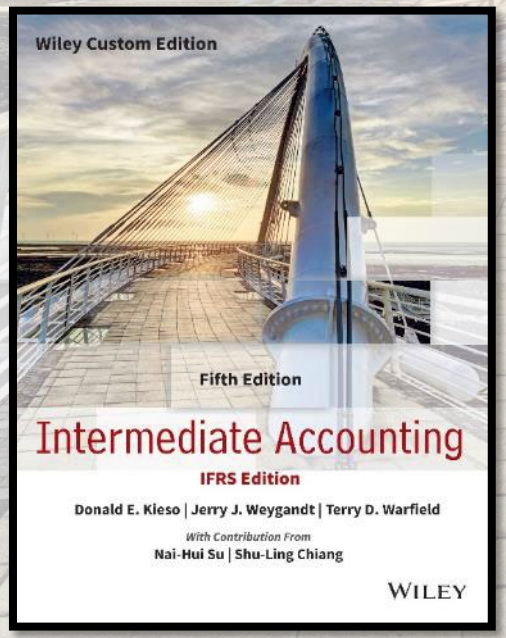
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Intermediate Accounting IFRS Edition

Kieso, Weygandt, Warfield

Fifth Edition

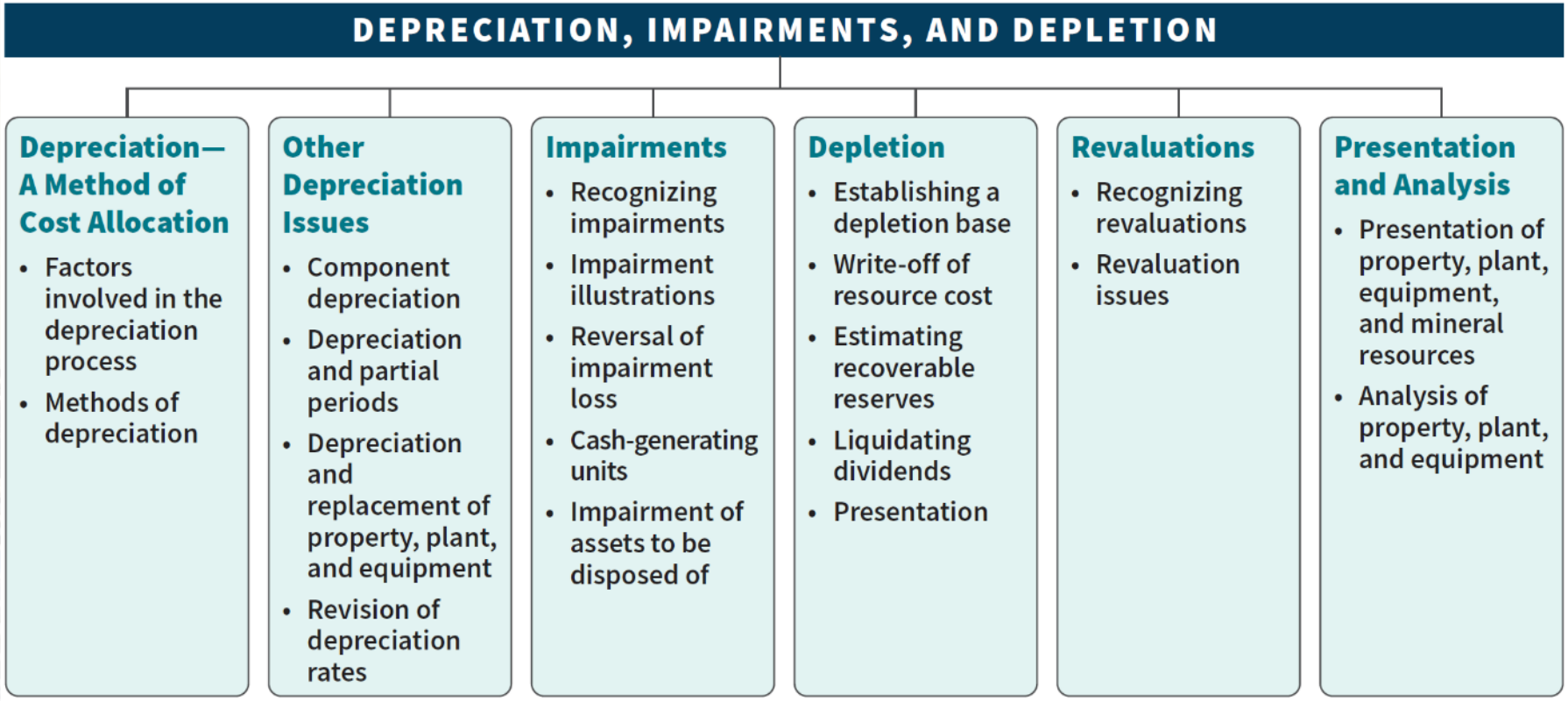
Chapter 8



Depreciation, Impairments, and Depletion

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PREVIEW OF CHAPTER 8



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Depreciation—A Method of Cost Allocation

- **Depreciation** is the accounting process of allocating the **cost of tangible assets to expense** in a systematic and rational manner to those periods expected to benefit from the use of the asset.
- Allocating costs of long-lived assets:
  - Fixed assets = Depreciation expense
  - Intangibles = Amortization expense
  - Mineral resources = Depletion expense

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Factors Involved in the Depreciation Process

Three basic questions:

1. What depreciable base is to be used?
2. What is the asset's useful life?
3. What method of cost apportionment is best?

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Factors Involved in Depreciation Process

Estimation of Service Lives

- Service life often differs from physical life.
- Companies retire assets for two reasons:
  1. **Physical factors** (casualty or expiration of physical life).
  2. **Economic factors** (**obsolescence**).

Physical Factors	Economic Factors
	
• Wear and tear, damage, and decay reduce service life. • Maintenance is important and can sustain and extend the service life.	• Technological changes in the industry cause an asset to become inadequate. • Changing company demands cause an asset to become obsolete.

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Depreciation—Method of Cost Allocation

Methods of Depreciation

The profession requires the method employed be **"systematic and rational."** Methods used include:

1. Activity method (units of use or production).
2. Straight-line method.
3. Diminishing (accelerated)-charge methods:
   a. Sum-of-the-years'-digits.
   b. Declining-balance method.

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Methods of Depreciation

Sum-of-the-Years' Digits Depreciation

Schedule

Year	Depreciable Base	Remaining Life in Years	Depreciation Fraction	Depreciation Expense	Book Value, End of Year
1	$450,000	5	5/15	$150,000	$350,000
2	450,000	4	4/15	120,000	230,000
3	450,000	3	3/15	90,000	140,000
4	450,000	2	2/15	60,000	80,000
5	450,000	1	1/15	30,000	50,000[a]
		15	15/15	$450,000	

[a]Residual value.

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Methods of Depreciation

Double-Declining Depreciation Schedule

Year	Book Value of Asset First of Year	Rate on Declining Balance[a]	Depreciation Expense	Balance Accumulated Depreciation	Book Value, End of Year
1	$500,000	40%	$200,000	$200,000	$300,000
2	300,000	40%	120,000	320,000	180,000
3	180,000	40%	72,000	392,000	108,000
4	108,000	40%	43,200	435,200	64,800
5	64,800	40%	14,800[b]	450,000	50,000

[a]Based on twice the straight-line rate of 20% ($90,000 ÷ $450,000 = .20; .20 × 2 = .40, or 40%).
[b]Limited to $14,800 because book value should not be less than residual value.

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Other Depreciation Issues

Component Depreciation

- **IFRS requires** that each part of an item of property, plant, and equipment that is significant to the total cost of the asset must be depreciated separately.

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Other Depreciation Issues

Depreciation and Partial Periods

How should companies compute **depreciation for partial periods?**

- Companies determine the **depreciation expense for the full year** and then
- **prorate** this depreciation expense **between the two periods** involved.

This process should continue throughout the useful life of the asset.

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Depreciation and Partial Periods

Accelerated Methods of Depreciation

	Sum-of-the-Years'-Digits	Double-Declining-Balance
1st full year	$(5/15 \times £10{,}000) = £3{,}333.33$	$(.40 \times £10{,}000) = £4{,}000$
2nd full year	$(4/15 \times 10{,}000) = 2{,}666.67$	$(.40 \times 6{,}000) = 2{,}400$
3rd full year	$(3/15 \times 10{,}000) = 2{,}000.00$	$(.40 \times 3{,}600) = 1{,}440$

Depreciation from July 1, 2026, to December 31, 2026

$6/12 \times £3{,}333.33 = £1{,}666.67$	$6/12 \times £4{,}000 = £2{,}000$

Depreciation for 2027

$6/12 \times £3{,}333.33 = £1{,}666.67$	$6/12 \times £4{,}000 = £2{,}000$
$6/12 \times 2{,}666.67 = 1{,}333.33$	$6/12 \times 2{,}400 = 1{,}200$
£3,000.00	£3,200
	or $(£10{,}000 - £2{,}000) \times .40 = £3{,}200$

Depreciation for 2028

$6/12 \times £2{,}666.67 = £1{,}333.33$	$6/12 \times £2{,}400 = £1{,}200$
$6/12 \times 2{,}000.00 = 1{,}000.00$	$6/12 \times 1{,}440 = 720$
£2,333.33	£1,920
	or $(£10{,}000 - £5{,}200) \times 0.40 = £1{,}920$

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Depreciation and Partial Periods

Fractional-Year Depreciation Policies

Machine purchased on June 10, 2026, for $45,000, no residual value, five-year estimated life.

Machine Cost = £45,000	Depreciation Allocated per Period Over 5-Year Life*					
Fractional-Year Policy	2026	2027	2028	2029	2030	2031
1. Nearest fraction of a year.	£5,000[a]	£9,000	£9,000	£9,000	£9,000	£4,000[b]
2. Nearest full month.	5,250[c]	9,000	9,000	9,000	9,000	3,750[d]
3. Half year in period of acquisition and disposal.	4,500	9,000	9,000	9,000	9,000	4,500
4. Full year in period of acquisition, none in period of disposal.	9,000	9,000	9,000	9,000	9,000	–0–
5. None in period of acquisition, full year in period of disposal.	–0–	9,000	9,000	9,000	9,000	9,000

[a] $6.667/12 \times £9,000$ [b] $5.333/12 \times £9,000$ [c] $7/12 \times £9,000$ [d] $5/12 \times £9,000$

*Rounded to nearest pound.

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Other Depreciation Issues

Depreciation and Replacement of PP&E

Does depreciation provide for the **replacement of assets?**

- Does not involve a current cash outflow.
- Funds for the replacement of the assets come from the revenues (generated through use of the asset).

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Other Depreciation Issues

Revision of Depreciation Rates

How should companies handle **revisions in depreciation rates?**

- Accounted for in the current and prospective periods
- Not handled retrospectively
- Not considered errors or extraordinary items

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Recognizing Impairments

- A long-lived tangible asset is **impaired** when a company is not able to recover the asset's carrying amount either through using it or by selling it.
- On an **annual basis, companies review the asset for indicators of impairments**—that is, a decline in the asset's cash-generating ability through use or sale.

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Impairment Test

If impairment indicators are present, then an **impairment test** must be conducted.

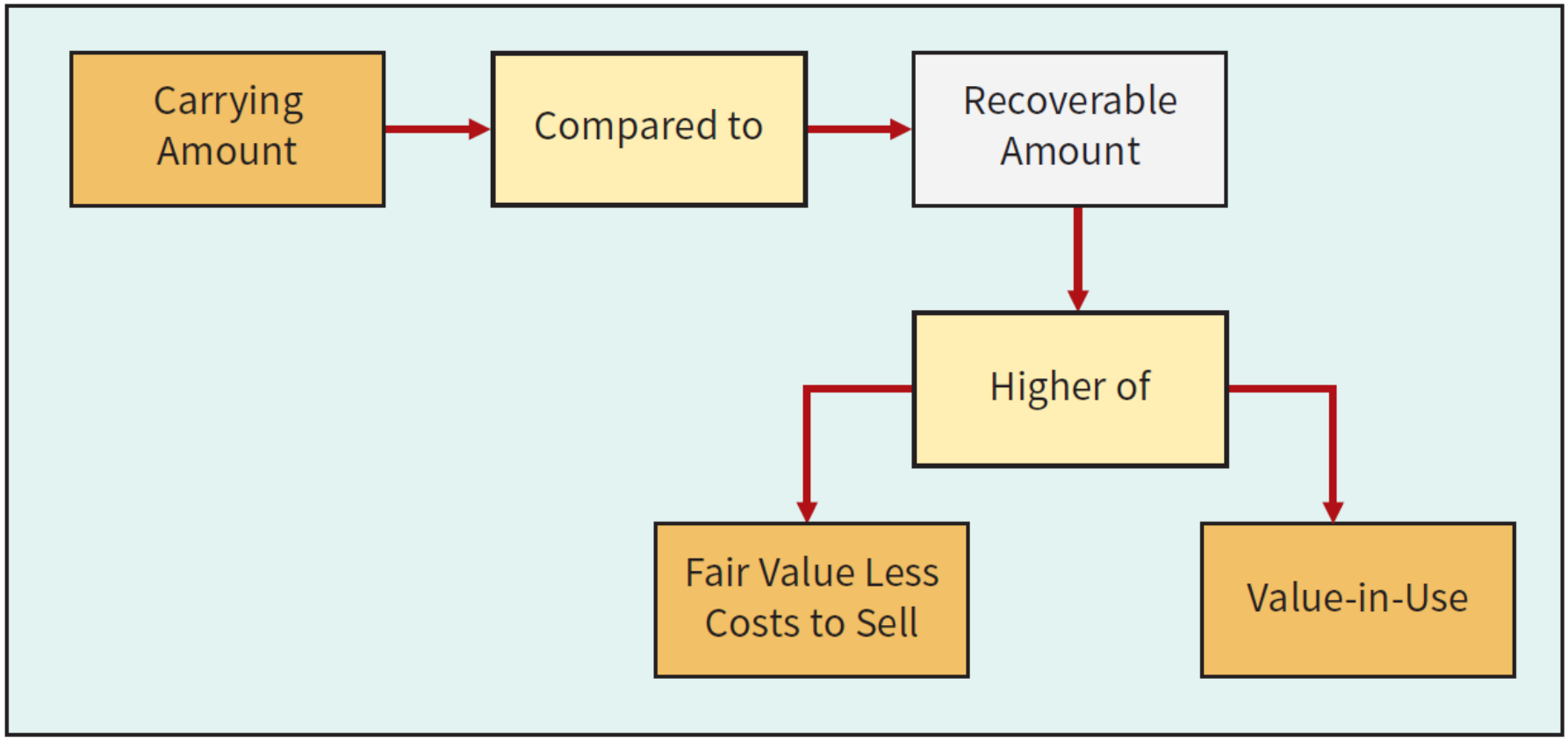


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Cash-Generating Units

- When it is not possible to assess a single asset for impairment because the single asset generates cash flows only in combination with other assets, companies **identify the smallest group of assets** that can be identified that generate cash flows independently of the cash flows from other assets.
- If the impairment test is based on a CGU, impairment should be recognized if the recoverable amount of a CGU is less than its carrying amount. An impairment loss attributable to a CGU should be written down the goodwill (if any) first until the carrying amount of the goodwill is reduced to zero; then, **the balance of the impairment loss will be allocated to the other assets in the CGU on a pro-rata basis, based on the carrying amount of each asset in the CGU.**

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Impairment of Assets to be Disposed Of

- Report the impaired asset at the **lower-of-cost-or-net realizable value** (fair value less costs to sell).
- No depreciation or amortization is taken on assets held for disposal during the period they are held.
- Can write up or down an asset held for disposal in future periods, as long as the carrying amount after the write up never exceeds the carrying amount of the asset before the impairment.

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Graphic of Accounting for Impairments

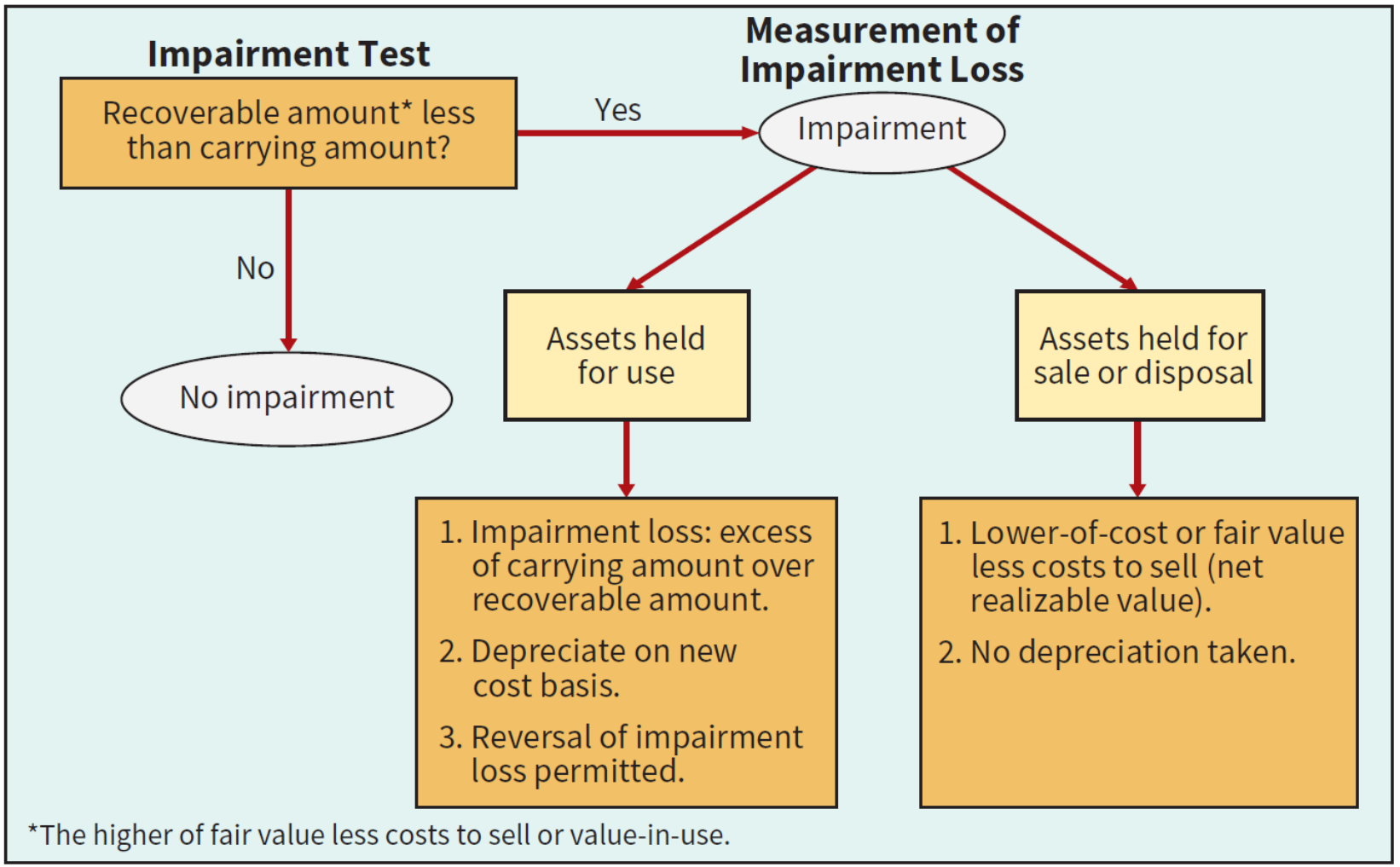


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Depletion

Natural resources can be divided into two categories:

1. **Biological assets** (timberlands)
   - Fair value approach (Chapter 9)
2. **Mineral resources** (oil, gas, and mineral mining).
   - Complete removal (consumption) of the asset.
   - Replacement of the asset only by an act of nature.

Depletion - process of allocating the cost of mineral resources.

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Establishing a Depletion Base

Computation of the depletion base involves:

1. Pre-exploratory costs.
2. Exploratory and evaluation costs.
3. Development costs.

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Write-off of Resource Cost

Normally, companies compute depletion on a **units-of-production method** (activity approach). Depletion is a function of the number of units extracted during the period. In this approach, the total cost of the mineral resource less residual value is divided by the number of units estimated to be in the resource deposit, to obtain a cost per unit of product. To compute depletion, the cost per unit is then multiplied by the number of units extracted.

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Estimating Recoverable Reserves

- Same as accounting for changes in estimates.
- Revise the depletion rate on a prospective basis.
- Divide the remaining cost by the new estimate of the recoverable reserves.

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Presentation on the Financial Statements

Disclosures related to E&E expenditures should include:

1. Accounting policies for exploration and evaluation expenditures, including the recognition of E&E assets.
2. Amounts of assets, liabilities, income and expense, and operating cash flow arising from the exploration for and evaluation of mineral resources.

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Recognizing Revaluations

Companies may **value long-lived tangible asset** subsequent to acquisition at **cost or fair value.**

Network Rail (GBR) elected to use fair values to account for its railroad network.

- Increased long-lived tangible assets by £4,289 million.
- Change in the fair value accounted for by adjusting the asset account and establishing an unrealized gain.
- Unrealized gain is often referred to as **revaluation surplus.**

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Revaluation Issues

Company can select to value only one class of assets, say buildings, and not revalue other assets such as land or equipment.

If a company selects only buildings,

- revaluation applies to all assets in that class of assets.
- A class of assets is a grouping of items that have a similar nature and use in a company's operations.
- Companies must also make every effort to keep the assets' values up to date.

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Presentation of Property, Plant, Equipment and Mineral Resources

- **Depreciating** assets, use Accumulated Depreciation.
- **Depleting assets** may include use of Accumulated Depletion account, or the direct reduction of asset.
- **Disclosures**
  - Basis of valuation (usually cost)
  - Pledges, liens, and other commitments

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Analysis of Property, Plant, and Equipment

Asset Turnover

Measures how efficiently a company uses its assets to generate sales.

Siemens Group
(in millions)

Net sales	€ 83,044
Total assets, 9/30/18	138,915
Total assets, 9/30/17	136,111
Net income	6,120

ILLUSTRATION 8.25

$$\text{Asset Turnover} = \frac{\text{Net Sales}}{\text{Average Total Assets}}$$

$$= \frac{€83,044}{(€138,915 + €136,111) \div 2}$$

$$= 0.60$$

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Analysis of Property, Plant, and Equipment

Asset Turnover

Measure of the ability to generate operating income from a particular level of sales.

Siemens Group
(in millions)

Net sales	€ 83,044
Total assets, 9/30/18	138,915
Total assets, 9/30/17	136,111
Net income	6,120

$$\text{Profit Margin on Sales} = \frac{\text{Net Income}}{\text{Net Sales}}$$

$$= \frac{€6,120}{€83,044}$$

$$= 7.4\%$$

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Analysis of Property, Plant, and Equipment

Asset Turnover

Measures a firm's success in using assets to generate earnings.

Siemens Group
(in millions)

Net sales	€ 83,044
Total assets, 9/30/18	138,915
Total assets, 9/30/17	136,111
Net income	6,120

Return on Assets = Profit Margin on Sales × Asset Turnover

$= .074 \times 0.60$

$= 4.4\%$

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Return on Assets

Rather than using the profit margin on sales, we can compute **return on assets** directly by dividing net income by average total assets. Using Siemens' data, the calculation is as follows.

$$\text{Return on Assets} = \frac{\text{Net Income}}{\text{Average Total Assets}}$$

$$= \frac{€6{,}120}{(€138{,}915 + €136{,}111) \div 2}$$

$$= 4.4\%$$

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Revaluation of Property, Plant, and Equipment (1 of 2)

The general rules for revaluation accounting are as follows.

1. When a company revalues its long-lived tangible assets above historical cost, it reports an unrealized gain that increases other comprehensive income.
2. If a company experiences a loss on revaluation (decrease of value below historical cost), the loss reduces income and retained earnings. Thus, gains on revaluation increase equity but not net income.

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Revaluation of Property, Plant, and Equipment (2 of 2)

3. If a revaluation increase reverses a decrease that was previously reported as a revaluation loss, a company credits the revaluation increase to income using the account Recovery of Revaluation increase above historical cost increases other comprehensive income and is credited to Unrealized Gain on Revaluation.
4. If a revaluation decrease reverses an increase that was reported as an unrealized gain, a company first reduces other comprehensive income by eliminating the unrealized gain. Any additional valuation decrease reduces net income and is reported as a loss on revaluation.

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Summary of Revaluation—2026

Date	Item	Land Fair Value	Retained Earnings	Accumulated Other Comprehensive Income (AOCI)
Jan. 1, 2026	Beginning balance	€400,000	€0	€ 0
Dec. 31, 2026	Revaluation	120,000	0	120,000
Dec. 31, 2026	**Ending balance**	**520,000**	**0**	**120,000**

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Summary of Revaluation—2027

Date	Item	Land Fair Value	Retained Earnings	Accumulated Other Comprehensive Income (AOCI)
Jan. 1, 2026	Beginning balance	€400,000	€ 0	€ 0
Dec. 31, 2026	Revaluation	120,000	0	120,000
Dec. 31, 2026	**Ending balance**	**520,000**	**0**	**120,000**
Jan. 1, 2027	Beginning balance	€520,000	€ 0	€ 120,000
Dec. 31, 2027	Revaluation	(140,000)	(20,000)	(120,000)
Dec. 31, 2027	**Ending balance**	**380,000**	**(20,000)**	**0**

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Summary of Revaluation—2028

Date	Item	Land Fair Value	Retained Earnings	Accumulated Other Comprehensive Income (AOCI)
Jan. 1, 2026	Beginning balance	€ 400,000	€ 0	€ 0
Dec. 31, 2026	Revaluation	120,000	0	120,000
Dec. 31, 2026	**Ending balance**	**520,000**	**0**	**120,000**
Jan. 1, 2027	Beginning balance	€ 520,000	€ 0	€120,000
Dec. 31, 2027	Revaluation	(140,000)	(20,000)	(120,000)
Dec. 31, 2027	**Ending balance**	**380,000**	**(20,000)**	**0**
Jan. 1, 2028	Beginning balance	€ 380,000	€(20,000)	€ 0
Dec. 31, 2028	Revaluation	35,000	20,000	15,000
Dec. 31, 2028	Ending balance	415,000	0	15,000

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- The purpose of this transfer is to **eliminate** the unrealized gain on the land that was sold.
- Transfers from Accumulated Other Comprehensive Income **cannot** increase net income.
- Even though the land has appreciated in value by €15,000, Unilever is **not able to recognize** this gain in net income over the periods that it held the land.

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Revaluation Summary—2026

Date	Item	Equipment Fair Value	Accumulated Depreciation	Retained Earnings	Accumulated Other Comprehensive Income (AOCI)
Jan. 1, 2026	Beginning balance	€1,000,000			€ 0
Dec. 31, 2026	Depreciation		€ 200,000	€(200,000)	
Dec. 31, 2026	Revaluation	(50,000)	(200,000)		150,000
Dec. 31, 2026	**Ending balance**	**950,000**	**0**	**(200,000)**	**150,000**

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Revaluation Summary—2027

Date	Item	Equipment Fair Value	Accumulated Depreciation	Retained Earnings	Accumulated Other Comprehensive Income (AOCI)
Jan. 1, 2026	Beginning balance	€1,000,000			€ 0
Dec. 31, 2026	Depreciation		€200,000	€(200,000)	
Dec. 31, 2026	Revaluation	(50,000)	(200,000)		150,000
Dec. 31, 2026	**Ending balance**	**950,000**	**0**	**(200,000)**	**150,000**
Jan. 1, 2027	Beginning balance	€ 950,000	€ 0	€(200,000)	€150,000
Dec. 31, 2027	Depreciation		237,500	(237,500)	
Dec. 31, 2027	Transfer from AOCI			37,500	(37,500)
Dec. 31, 2027	Revaluation	(380,000)	(237,500)	(30,000)	(112,500)
Dec. 31, 2027	**Ending balance**	**570,000**	**0**	**(430,000)**	**0**

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Revaluation Summary—2028

Date	Item	Equipment Fair Value	Accumulated Depreciation	Retained Earnings	Accumulated Other Comprehensive Income (AOCI)
Jan. 1, 2026	Beginning balance	€1,000,000			€ 0
Dec. 31, 2026	Depreciation		€200,000	€(200,000)	
Dec. 31, 2026	Revaluation	(50,000)	(200,000)		150,000
Dec. 31, 2026	**Ending balance**	**950,000**	**0**	**(200,000)**	**150,000**
Jan. 1, 2027	Beginning balance	€ 950,000	€ 0	€(200,000)	€150,000
Dec. 31, 2027	Depreciation		237,500	(237,500)	
Dec. 31, 2027	Transfer from AOCI			37,500	(37,500)
Dec. 31, 2027	Revaluation	(380,000)	(237,500)	(30,000)	(112,500)
Dec. 31, 2027	**Ending balance**	**570,000**	**0**	**(430,000)**	**0**
Jan. 1, 2028	Beginning balance	€ 570,000	€ 0	€(430,000)	€ 0
Dec. 31, 2028	Depreciation		190,000	(190,000)	
Dec. 31, 2028	Revaluation	(120,000)	(190,000)	20,000	50,000
Dec. 31, 2028	**Ending balance**	**450,000**	**0**	**(600,000)**	**50,000**

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